

AI UNLEASHES CHAOS

Lorenz attractor -- circuit by Paul Horowitz

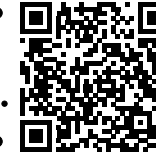
PCB by Jason Gallicchio and Claude Opus 5 Max | 2026-09-15 rev A



$$\begin{aligned}dx/dt &= s(y - x) \\ dy/dt &= r x - y - x z \\ dz/dt &= x y - b z\end{aligned}$$

$s = 10$ $b = 8/3$
 $r = 21$ to 37 on the knob

put x and z on 'scope X-Y to see this



this project, and how
it was made
[github.com/gallicchio
/aLunleashes_chaos](https://github.com/gallicchio/aLunleashes_chaos)

- Every term weighs 1 MEG / R.
s is $R1 = R2 = 100k$, and b is $R7 = 374k$.
r is $R3 = 27k$ plus the knob RV1, which sweeps r from 21 to 37.
The MPY634s form $A*B/10$, and both minus signs come from swapped inputs.
- Signals are 0.1 V per unit, and the time scale is $\tau = 1 \text{ MEG} \times C$.



Paul's original circuit
[seti.harvard.edu/unusual_stuff
/misc/lorenz.htm](https://seti.harvard.edu/unusual_stuff/misc/lorenz.htm)

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